

Usaid Siddiqui

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Education

The University of Alabama

May 2028

BS in Computer Science & Electrical Engineering, Minor in Math & Robotics

GPA: 4.00/4.00

- National Merit Scholar, Honors College, President's List (Fall 2024 – Spring 2026), Association for Computing Machinery

Skills

Languages: Python, SQL (MySQL), PHP, JavaScript, C, C++, Bash, VHDL, Assembly

Web & Data: WordPress, REST APIs, HTML/CSS, Pandas, NumPy, Excel

Frameworks: ROS/ROS2, PyTorch, TensorFlow, Scikit-Learn

Embedded & Hardware: MPLAB X IDE, KiCAD, EasyEDA, OrCAD PSpice, serial/sensor interfacing

CAD/Sim & Cloud: Webots, Fusion360, SolidWorks, AutoCAD, AWS Cloud Practitioner

Experience

Software Engineer Intern, Alabama Productivity Center – Tuscaloosa, AL

July 2026 – Aug 2026

- Digitized a manual spreadsheet-based intake process for the Alabama Center for Real Estate by engineering a full-stack WordPress/PHP application with an admin dashboard providing inline status editing, searchable filtering, internal notes, and CSV export
- Enabled non-technical staff to maintain the platform independently by building a no-code interface to add, remove, and toggle the requirement of form fields, extending its usable lifespan without developer support
- Hardened handling of student PII by architecting a non-public data model with capability- and nonce-gated file storage served outside the public web root, preventing unauthenticated access and URL enumeration

Machine Learning Research Assistant, Particle Physics Lab – Tuscaloosa, AL

Aug 2025 – Present

- Improved feature quality on unlabeled particle physics data by designing and training a self-supervised Vision Transformer, achieving representations that generalize better than supervised baselines in low-data regimes
- Reduced training instability on distributed GPU clusters by implementing Mixed Precision with gradient scaling, maximizing hardware utilization
- Evaluated model performance through linear probing and exploratory data analysis to assess representation quality and downstream classification transferability

Robotics Research Assistant, Trustworthy Autonomy Lab – Tuscaloosa, AL

April 2026 – Present

- Developed and deployed a custom TurtleBot3 ROS 2 simulation package integrating Webots with the TurtleBot3 Burger, enabling networked node, SLAM, and localization testing without physical hardware access
- Built a data pipeline converting real-world SLAM sensor output into simulation environments by translating occupancy grid data into scaled 3D geometry, accurately recreating the physical lab space
- Interfaced distributed ROS 2 nodes over a local network to stream and visualize live sensor telemetry from mobile robot hardware

Math Tutor, Mathematics Technology Learning Center – Tuscaloosa, AL

Aug 2025 – Present

- Provided individual and group tutoring in Calculus 1–3 and Linear Algebra, assisting 20+ students with prerequisite review and exam preparation

Avionics Engineer, Alabama Rocketry Association – Tuscaloosa, AL

Aug 2024 – Dec 2025

- Wrote embedded C++ firmware to sample, timestamp, and log multi-sensor flight data for post-flight analysis
- Designed and tested flight PCBs using EasyEDA to ensure stable avionics performance under launch conditions
- Validated electronics under simulated launch stress to ensure reliable data collection and recording